

along with strong analytical and problem-solving skills, communication skills and the ability to excel in a team environment.

"Only those electronics engineers who are strong in electronics design fundamentals, and have mathematical and analytical aptitude, coupled with an interest in design and verification, can grow into good VLSI talent," says Aghoramoorthy.

Other than IITs, NITs, IIITs and other engineering colleges which include VLSI design in their engineering curriculum, there are technical training institutes like EFY Tech Center which impart dedicated VLSI training as per the current industry requirements.

What does the industry look for?

While recruiting, companies not only look for academic performance but also for the knowledge gained by the candidates at the end of their engineering studies.

Deepak D. Agarwal, senior marketing executive, Sandeepani School of Embedded System Design, affirms, "Everyday we see a new technology popping up in the market in response to the market demand. Due to this, the gap between the academics (which is more of exam-centric teaching) and the industry (which rides on the quality-manpower to address market requirements) has widened over a period of time, as it is practically impossible to keep on changing the engineering syllabus every now and then to meet the industry requirements."

He adds, "Finishing schools help freshers gain the much needed industry exposure and practice to make them employable. On the other hand, exam-centric teaching and preparation methodology makes the students lose their grip over the knowledge and they just study for getting good grades." Hence many institutes these days are increasingly focusing on the development of well-structured, industry-driven academic programmes that benefit both the students as well as the

electronics industry via initiatives such as establishment of incubation centres to help start-up companies.

Available career options

Some good segments include industrial and automotive microcontrollers, wireless infrastructure, consumer digital TV and handheld devices, and computing/storage infrastructure. Emerging areas include smart-energy product development, medical telemetry application, and high-end application processor development.

Freshers are initially given block-level and verification tasks. However, with experience and talent, they get to work on more challenging tasks. For example, while front-end engineers get into design and architectural engineering, back-end engineers get into automation and full-chip tasks.

Once an engineer has gained enough hands-on experience, he can either choose to grow as a team leader, then project leader and project manager, or become an individual contributor working on tasks like methodology development and R&D. You can climb the ladder through such positions as member of technical staff, senior member of technical staff, principal member of technical staff and chip architect. The various positions that you can apply for include design engineer, product engineer, test engineer, application engineer, process engineer, packaging engineer and CAD engineer as per your area of interest.

Kathuria adds: "There are a number of areas to choose from. One such area is reconfigurable computing. It is an interesting and pretty recent development in microelectronics that involves fabricating circuits which can be reprogrammed on the fly! And no, we are not talking about microcontrollers running with EEPROM inside. Reconfigurable computing involves specially fabricated FPGAs that when programmed act just like normal electronic circuits. These are designed such that by changing or reprogramming the connections between numerous

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sub-modules, the FPGAs can be made to behave like any circuit you wish."

Opportunities and desired skillset

To enter, survive and grow in this competitive industry, you must have the specialised skillset and qualification along with right aptitude and a desire to continuously learn and evolve. You should be willing to put extra efforts to keep abreast of the latest technology trends that are shaping the world.

Major recruiters in this field are Texas Instruments, PMC Sierra, Infineon, Alliance Semiconductor, Freescale Semiconductor, Analog Devices, Cadence, Synopsys, Mentor Graphics, HCL, Intel, Lucent, Micron Tech, National Semiconductor, Motorola, Philips Semiconductor, Qualcomm, Sarsen, Atrienta, Conexant, Moschip, Cradle Tech, Synplicity, Wipro, TCS